

What local Selmer conditions are for

Seven things the formalism buys, the theorems that illustrate each, and where the involution theorem sits among them

companion to selmer-involution.typ

1. The formalism, and the one idea in it

A Selmer group is never attached to an object. It is attached to a **pair**: a Galois module and a choice of local subspaces. Write M for a finite $G_{\mathbb{Q}}$ -module (or a p -adic representation), and for each place v choose a subgroup

$$L_v \subseteq H^1(\mathbb{Q}_v, M).$$

The collection $\mathcal{L} = \{L_v\}$ is a **Selmer structure**, subject to one finiteness axiom: $L_v = H_{\text{ur}}^1(\mathbb{Q}_v, M)$ for all but finitely many v . The associated Selmer group is

$$\text{Sel}(M, \mathcal{L}) = \ker \left(H^1(\mathbb{Q}, M) \longrightarrow \prod_v H^1(\mathbb{Q}_v, M)/L_v \right).$$

That is the whole definition. The vocabulary is Mazur–Rubin’s; the objects are older.

The one idea. A global object — a class in $H^1(\mathbb{Q}, M)$, ultimately a rational point or an element of III — is pinned down by a **finite list of independent local questions**. Nothing in the definition remembers where M came from. Two entirely different arithmetic problems that produce the same $(M, \mathcal{L})$ have literally the same Selmer group, and two problems that produce the same M with different $\mathcal{L}$ differ by an amount that is **computable place by place**.

Everything below is a consequence of that sentence. The classical case is $M = E[n]$ and $L_v = \text{im}(E(\mathbb{Q}_v)/nE(\mathbb{Q}_v) \rightarrow H^1(\mathbb{Q}_v, E[n]))$, giving $\text{Sel}_n(E)$; but the point of the formalism is that this is one point in a space of choices, and that **moving** the choice is the technique.

Three structural facts make the formalism work, and each is the source of a family of applications.

(F1) Finiteness. $\text{Sel}(M, \mathcal{L})$ is finite and effectively computable: classes are unramified outside a finite set S , and $H_S^1(\mathbb{Q}, M)$ is finite by Hermite–Minkowski. Each local condition is a computation over one local field.

(F2) Duality. If $M^* = \text{Hom}(M, \mu_n)$ is the Cartier dual, Tate local duality makes $H^1(\mathbb{Q}_v, M) \times H^1(\mathbb{Q}_v, M^*) \rightarrow \mathbb{Q}/\mathbb{Z}$ a perfect pairing, and $\mathcal{L}^\perp = \{L_v^\perp\}$ is a Selmer structure for M^* . When M is **self-dual** — as $E[n]$ is, by the Weil pairing — the local conditions live in a metabolic space and the Kummer images L_v are **maximal isotropic**.

(F3) Functoriality. A morphism of modules carries Selmer structures to Selmer structures. This is what lets a statement about one curve become a statement about another.

The first thing to notice is that (F2) already dictates the **size** of the classical local condition with no reference to the curve — which is Lemma B’ of the companion note, and the reason the comparison there is only ever about **which** subspace, never about how big.

2. Application 1: descent is finite because the conditions are local

The oldest application, and the one everything else is built on. Mordell–Weil gives no algorithm; the exact sequence

$$0 \rightarrow E(\mathbb{Q})/nE(\mathbb{Q}) \rightarrow \mathrm{Sel}_n(E) \rightarrow \mathrm{III}(E)[n] \rightarrow 0$$

does, because the middle term is cut out by conditions that can each be checked over a single $\mathbb{Q}_v$. **Rank bounds are computable; ranks are not.** Every `ellrank`, every `mwrnk`, every 2-descent in the companion note is this observation.

What is worth saying is how far the consequence reaches. Because Sel_n is defined by local conditions, it is amenable to **counting** in a way $E(\mathbb{Q})$ is not: a Selmer element is a local datum, and local data can be parametrised by lattice points in a fundamental domain.

Theorem (Bhargava–Shankar, [arXiv:1006.1002](#), *Annals* **181** (2015) 191–242). When elliptic curves over $\mathbb{Q}$ are ordered by height, the average size of $\mathrm{Sel}_2(E)$ is exactly 3. Consequently the average rank is at most $3/2$.

The mechanism is exactly (F1): a 2-Selmer element is a binary quartic form with prescribed invariants and local solubility everywhere, so the average is a volume computation with a congruence condition at each place. The same method gives average $\mathrm{Sel}_3 = 4$, $\mathrm{Sel}_5 = 6$ ([arXiv:1312.7859](#)), average rank < 0.885 , and a positive proportion of curves of rank 0 and of rank 1 satisfying Birch–Swinnerton-Dyer. None of this is available for $E(\mathbb{Q})$ directly. It is available for Sel_n **because** the definition is a list of local conditions.

3. Application 2: the comparison formula

If (F1) is why Selmer groups are computable, the following is why they are **usable**. It is the single most-quoted consequence of the formalism, and it is a global statement whose right-hand side is entirely local.

Theorem (Greenberg–Wiles). Let M be a finite $G_{\mathbb{Q}}$ -module, M^* its Cartier dual, $\mathcal{L}$ a Selmer structure and $\mathcal{L}^{\perp}$ its dual. Then

$$\frac{\#\mathrm{Sel}(M, \mathcal{L})}{\#\mathrm{Sel}(M^*, \mathcal{L}^{\perp})} = \frac{\#H^0(\mathbb{Q}, M)}{\#H^0(\mathbb{Q}, M^*)} \cdot \prod_v \frac{\#L_v}{\#H^0(\mathbb{Q}_v, M)},$$

the product over all places, almost all factors being 1.

Read it as: **the difference between a Selmer group and its dual is a sum of local terms**. One does not need to know either side to know their ratio. Four uses, in increasing order of ambition.

3.1. Descent by isogeny

Take $\varphi : E \rightarrow E'$ an isogeny with dual $\hat{\varphi}$, and $M = E[\varphi]$, so $M^* = E'[\hat{\varphi}]$ by the Weil pairing. With the Kummer local conditions on both sides, Greenberg–Wiles becomes the classical formula of Cassels comparing $\mathrm{Sel}^{\varphi}(E)$ with $\mathrm{Sel}^{\hat{\varphi}}(E')$ as a product of local indices $\#(E'(\mathbb{Q}_v)/\varphi E(\mathbb{Q}_v))$. This is how descent by 2-isogeny is **organised**: one computes each local index — a Tamagawa-number-sized quantity — and reads off the difference of the two Selmer ranks. It is the route to the classical results on the congruent number curve $y^2 = x^3 - n^2x$, where the local indices are the entries of Monsky’s matrix and the whole 2-descent becomes a rank computation over $\mathbb{F}_2$.

3.2. Parity, without computing either side

When M is self-dual and $\mathcal{L}, \mathcal{L}'$ are both self-dual Selmer structures, the formula degenerates to the congruence used at the end of the companion note:

$$\dim \mathrm{Sel}(\mathcal{L}) \equiv \dim \mathrm{Sel}(\mathcal{L}') + \sum_v \dim(L_v/(L_v \cap L'_v)) \pmod{2}.$$

Two maximal isotropic subspaces of a metabolic space meet in something of predictable parity; that is all this is. It is weak — it sees only the sum mod 2 — but it is **unconditional and local**, and it is the reason parity results are so much easier than rank results (Section 5).

3.3. Changing the condition at one place

This is the operational heart of the subject. Suppose $\mathcal{L} \subseteq \mathcal{L}'$ differ only at a single place q . Poitou–Tate global duality gives an exact sequence

$$0 \rightarrow \mathrm{Sel}(M, \mathcal{L}) \rightarrow \mathrm{Sel}(M, \mathcal{L}') \rightarrow L'_q/L_q \rightarrow \mathrm{Sel}(M^*, \mathcal{L}^\perp)^\vee \rightarrow \mathrm{Sel}(M^*, \mathcal{L}'^\perp)^\vee \rightarrow 0.$$

What this is for. Relaxing the condition at q can only **grow** the Selmer group, and it grows by an amount governed by the **dual** Selmer group. So: if you can produce a global class that is locally trivial everywhere except at q , where it is non-trivial, the sequence forces the dual Selmer group to **shrink**. That is the entire strategy of Euler systems, of Ribet’s converse to Herbrand, and of level-raising. One manufactures global classes with prescribed local behaviour, and duality converts them into upper bounds.

3.4. The class-group face

Take $M = \mu_p$ and $M^* = \mathbb{Z}/p$ with everywhere-unramified conditions. Then $\mathrm{Sel}(\mu_p, \mathrm{ur})$ is the p -Selmer group of the ring of integers — units modulo p -th powers, extended by $\mathrm{Cl}(K)[p]$ — while $\mathrm{Sel}(\mathbb{Z}/p, \mathrm{ur})$ is $\mathrm{Hom}(\mathrm{Cl}(K), \mathbb{Z}/p)$, the everywhere-unramified $\mathbb{Z}/p$ -extensions. Greenberg–Wiles then relates the p -rank of the class group to the p -rank of the unit group, with the local terms sitting at $v \mid p$ and $v \mid \infty$; the archimedean factors are Dirichlet’s unit theorem in disguise, and the resulting inequalities are the reflection principles of Scholz and Leopoldt. It is worth doing once: the same formula that runs 2-descent also runs genus theory.

4. Application 3: bounding III — Euler systems

Selmer groups are upper bounds for ranks; the hard direction is bounding them from **above**, which means proving III is small. The only known general mechanism is Section 3.3, applied with global classes that come from geometry.

Kolyvagin’s theorem. If $E/\mathbb{Q}$ is modular and the Heegner point y_K has infinite order, then $\mathrm{rank} E(\mathbb{Q}) = 1$ and $\mathrm{III}(E)$ is finite. Combined with Gross–Zagier: if $\mathrm{ord}_{s=1} L(E, s) \leq 1$ then the analytic rank equals the algebraic rank and III is finite.

The proof is Section 3.3 run infinitely often. Kolyvagin builds, from Heegner points over ring class fields, a system of **derived classes** $c(n) \in H^1(K, E[p^k])$ indexed by squarefree products of “Kolyvagin primes” ℓ , each of which is **locally in the Selmer condition at every place except at the $\ell \mid n$** , where its localisation is computable and non-zero. Feeding these into the duality sequence one place at a time bounds the Selmer group by the index of the Heegner point. The primes ℓ are chosen by Chebotarev precisely so that the local condition at ℓ **can be moved**.

The same architecture, with different classes:

- **Rubin** ([Invent. 103 \(1991\)](#)): elliptic units, giving the main conjecture of Iwasawa theory for imaginary quadratic fields, hence the finiteness of III for CM curves of analytic rank 0 — this is the input the note on III at the Fermat primes uses.
- **Kato**: Beilinson elements in K_2 of modular curves, giving that $L(E, 1) \neq 0 \Rightarrow \mathrm{Sel}_{p^\infty}(E)$ finite, for all modular E and almost all p .
- **Skinner–Urban**: the converse divisibility, via Eisenstein congruences on $\mathrm{GU}(2, 2)$; with Kato this yields the p -part of BSD in analytic rank 0 for a large class of curves.

In all four cases the theorem being proved is about L -functions and rational points; the **mechanism** is the movement of one local condition at a time, and the duality sequence that converts a local non-vanishing into a global bound.

5. Application 4: parity is local

Because the Kummer conditions are maximal isotropic (F2), the **parity** of a Selmer rank is a sum of local invariants, and can be computed with no global information at all. This is the cheapest thing local conditions give, and the source of some of the most quotable theorems.

p -parity theorem (Nekovář; Dokchitser–Dokchitser, [Invent. 178 \(2009\)](#)). For $E/\mathbb{Q}$ and any prime p ,

$$(-1)^{\dim \mathrm{Sel}_{p^\infty}(E)} = w(E),$$

the global root number, itself a product of local root numbers w_v .

So the parity half of Birch–Swinnerton-Dyer is a **theorem**, and it is a theorem because both sides factor over places. The Dokchitser’s technique — **regulator constants** — is worth knowing as a method: one compares a Selmer group over $\mathbb{Q}$ with its behaviour in a G -extension, and Brauer relations among the subgroups of G turn the comparison into a product of local terms. The by-product, “BSD modulo squares” ([Annals 172 \(2010\)](#)), constrains the Birch–Swinnerton-Dyer quotient by purely local data.

An older, more elementary instance of the same idea, and the one closest to the companion note:

Kramer ([Trans. AMS 264 \(1981\)](#)). For $E/\mathbb{Q}$ and $K = \mathbb{Q}(\sqrt{d})$, $\dim \mathrm{Sel}_2(E/K)$ is expressed through local norm indices $E(\mathbb{Q}_v)/N_{K_w/\mathbb{Q}_v} E(K_w)$; in particular the 2-Selmer ranks of E and its quadratic twist E^d are related by an explicit sum of local terms.

That is the general theorem of which the situation in the companion note is a very rigid special case: there, the twist is by -1 , ramified only at 2 and ∞ , and the local terms are shown to vanish one by one rather than merely in aggregate.

6. Application 5: statistics — local conditions as random Lagrangians

If a Selmer group is the intersection of a global subspace with a product of local Lagrangians, then **modelling the local conditions as random** should predict Selmer statistics. It does, spectacularly.

Poonen–Rains ([arXiv:1009.0287](#), [JAMS 25 \(2012\) 245–269](#)). The p -Selmer group of E is naturally the intersection of a discrete maximal isotropic subspace with a compact open maximal isotropic subspace in a locally compact quadratic space over $\mathbb{F}_p$. Modelling the first as random predicts that the m -th moment of $\#\mathrm{Sel}_p$ is $\prod_{i=1}^m (p^i + 1)$; in particular the average is $p + 1$.

For $p = 2$ that is the average 3 of Bhargava–Shankar — proved independently, and matching. The model also reproduces Heath-Brown’s and Kane’s theorems on 2-Selmer distributions in quadratic twist families. **The heuristic is a heuristic about local conditions and nothing else.**

Once one thinks this way, the behaviour of Selmer ranks **along a twist family** becomes a question about how the Lagrangians move.

Klagsbrun–Mazur–Rubin ([Annals 178 \(2013\) 287–320](#) [Compositio 150 \(2014\) 1077–1106](#)). In the family of quadratic twists of a fixed E/K , the proportion of twists with **even** 2-Selmer rank exists and

equals an explicit product of local factors — the **disparity** $\delta(E, K)$; and the distribution of 2-Selmer ranks is governed by a **Markov chain**, in which adjoining one more prime to the twisting parameter moves the rank by ± 1 with computable probabilities.

This is the frame the companion note’s even- a table belongs in. Adjoining the prime 2 to the twisting parameter moves one Lagrangian at one place, and a single moved place changes the Selmer rank by exactly 0 or 1. The observed “14 of 16 even a differ by exactly 1” is not a curiosity: it is the generic outcome of a single-place discrepancy, and its **absence** for odd a is what the theorem there proves.

The deepest current result in this direction removes the heuristic entirely.

A. Smith ([arXiv:1702.02325](https://arxiv.org/abs/1702.02325)). Let $E/\mathbb{Q}$ have full rational 2-torsion and no rational cyclic subgroup of order 4. Then the 2^∞ -Selmer groups of the quadratic twists of E are distributed as Delaunay’s heuristic predicts; in particular 100% of twists have rank 0 or 1 (Goldfeld’s conjecture for the family), and the number with rank ≥ 2 among $|d| < N$ is $o(N)$.

Corollary. The congruent numbers of the form 1, 2, 3 modulo 8 have natural density zero.

A statement about right triangles with rational sides, proved by controlling how local conditions at the primes of d interact as d varies. If one wants a single answer to “why are local Selmer conditions useful”, this is a good one.

7. Application 6: local conditions as definitions — deformations and Bloch–Kato

So far the local conditions came from a curve. The formalism’s real reach is that one may **impose** them, and thereby define objects that have no elementary description.

7.1. Deformation theory

Let $\bar{\rho} : G_{\mathbb{Q}} \rightarrow \mathrm{GL}_2(\mathbb{F}_p)$ be residually irreducible. The tangent space to the functor of deformations of $\bar{\rho}$ with prescribed behaviour at each place — unramified outside Σ , ordinary at p , of fixed type at $\ell \in \Sigma$ — is a **Selmer group**

$$H_{\mathcal{L}_{\Sigma}}^1(\mathbb{Q}, \mathrm{ad}^0 \bar{\rho}),$$

where $\mathcal{L}_{\Sigma}$ is precisely the list of local deformation conditions. Wiles’s numerical criterion for $R = \mathbb{T}$ compares two such Selmer groups differing at finitely many places, and the comparison is made by Greenberg–Wiles (Section 3): the local terms are exactly the Euler factors that appear in the change of level. **Level-raising and level-lowering are changes of local condition.** Ribet’s converse to Herbrand’s theorem is the same move in the abelian case: a non-zero Selmer class is an unramified extension, and it is constructed by congruence.

7.2. Bloch–Kato

For a p -adic representation V , there is no Kummer map to define L_v with. Bloch and Kato supply one:

$$H_f^1(\mathbb{Q}_p, V) = \ker(H^1(\mathbb{Q}_p, V) \rightarrow H^1(\mathbb{Q}_p, V \otimes B_{\mathrm{cris}})),$$

with H_{ur}^1 at $v \neq p$, and variants $H_e^1 \subseteq H_f^1 \subseteq H_g^1$. The resulting $\mathrm{Sel}_f(V)$ is **conjecturally** the object whose dimension is the order of vanishing of the L -function of the motive. Every modern formulation of a Birch–Swinnerton-Dyer-type conjecture is a statement about a Selmer group defined this way.

Mazur and Rubin remark, in the paper of Section 9, that most of what they do “applies more generally to the Bloch–Kato Selmer groups attached to a motive”, and that “**the subtlest problem in the general case, as in the case of elliptic curves, is to understand the local condition at p** ”. That is worth underlining, since it is exactly where the companion note’s proof spends all of its effort: at $v = 2$, for $p = 2$.

8. Application 7: consequences that do not look like Selmer theory

Two results where the local conditions are the **tool** and the conclusion is in another subject entirely.

Mazur–Rubin ([Invent. 181 \(2010\) 541–575](#)). By controlling how the 2-Selmer rank changes when the twisting parameter acquires one more prime, one produces, over suitable number fields K , quadratic twists of prescribed 2-Selmer rank — in particular many twists with $\text{Sel}_2 = 0$, hence rank 0 and trivial Mordell–Weil group.

Consequence. Hilbert’s tenth problem has a negative answer over the ring of integers of every number field K admitting such a curve: an elliptic curve of rank 1 over $\mathbb{Q}$ that keeps rank 1 over K gives a diophantine model of $\mathbb{Z}$ in $\mathcal{O}_K$.

The undecidability of a class of diophantine problems, obtained by moving a local condition at a finite prime. The companion result, **Diophantine stability** ([Mazur–Rubin, 2018](#)), shows in the same way that $E(K) = E(\mathbb{Q})$ for many extensions K — rank does not grow — which is again a statement that certain local conditions do not move.

9. Where the involution theorem sits: Selmer companions

The companion note proves that two **non-isogenous** curves have **equal** 2-Selmer groups, and it does so by the exact route (F3) predicts: identify the modules, then compare local conditions. There is a literature that names this phenomenon, and the note’s theorem turns out to sit just outside it.

Definition (Mazur–Rubin, [arXiv:1203.0620](#)). Two elliptic curves E_1, E_2 over K are **n -Selmer companions** if for every quadratic character χ of K there is an isomorphism $\text{Sel}_n(E_1^\chi/K) \cong \text{Sel}_n(E_2^\chi/K)$.

Their sufficient criterion is the exact analogue of the note’s Theorem A plus $(\star)$, with the local work concentrated at the potentially multiplicative places:

Theorem 3.1 (Mazur–Rubin). Let S_i be the set of primes where E_i has potentially multiplicative reduction, and let $m = p^{k+1}$ if $p \leq 3$, $m = p^k$ if $p > 3$. Suppose

- (i) there is a G_K -isomorphism $E_1[m] \cong E_2[m]$;
- (ii) $S_1 = S_2$;
- (iii) for all $\mathfrak{l} \in S_1$, the isomorphism sends $C_{E_1/K_1}[m]$ to $C_{E_2/K_1}[m]$;
- (iv) for every $\mathfrak{p} \mid p$, either $\mathfrak{p} \in S_1$, or $k = 1$, both curves have good reduction at $\mathfrak{p}$, and $e(\mathfrak{p}/p) < p - 1$.

Then $\text{Sel}_{p^k}(E_1^\chi/F) \cong \text{Sel}_{p^k}(E_2^\chi/F)$ for every finite F/K and every χ — and in fact the two groups are **equal** inside the common $H^1(F, E_1^\chi[p^k])$.

Here $C_{E/K_1}[m] = \tau_{E/K_1}(\mu_m)$ is the **canonical subgroup** cut out by the Tate parametrisation — the same object the survey calls $C_{\text{can}(v)}$, and Mazur–Rubin’s Lemma 2.5 (if $p \nmid \text{ord}_1(j(E_1))$, every G -isomorphism $E_1[p^k] \cong E_2[p^k]$ automatically matches the canonical subgroups) is the structural reason the survey’s recipe can read the local condition off reduction data alone.

9.1. The criterion does not reach $y^2 = x^3 + a$

Apply Theorem 3.1 with $p = 2$, $k = 1$ to E_a and $E_{-1/a}$. Both have $j = 0$, hence **potentially good reduction everywhere**, so $S_1 = S_2 = \emptyset$ and hypothesis (ii) holds vacuously, (iii) is empty — and (iv) **fails**: at $\mathfrak{p} = 2$ we are not in S_1 , and the alternative demands $e(\mathfrak{p}/2) < p - 1 = 1$, which no place satisfies. Independently, (i) would demand $E_a[4] \cong E_{-1/a}[4]$, whereas Theorem A supplies the isomorphism only on $E[2]$.

So the theorem of the companion note is a companion-type result for a pair the known criterion cannot reach, and it fails to reach it at exactly the place where the note's proof does all its work. The exhaustive check over $a \bmod 8$ is not a shortcut around a general argument; it is standing in for the one hypothesis the literature also cannot dispose of.

9.2. What the note actually proves, in this vocabulary

Twisting E_a by d gives E_{ad^3} , and the involution commutes with it: $-1/(ad^3) \equiv (-1/a)d^3$ modulo sixth powers. So the note's theorem, applied to the odd integer ad^3 , says:

Observation. For a odd, E_a and $E_{-1/a}$ are 2-Selmer companions **along every quadratic character unramified at 2**: for every odd squarefree d ,

$$\text{Sel}_2(E_a^{(d)}) = \text{Sel}_2(E_{-1/a}^{(d)})$$

as subgroups of one $H^1(\mathbb{Q}, M)$. The even- a table is the statement that companionship **fails** for χ ramified at 2, and the failure is by the single step ± 1 that one moved place permits.

That is a slightly stronger and more standard-sounding form of the same theorem, and it says precisely which characters are allowed. It also explains why the note's Theorem B is the right structural remark: the involution is the quadratic twist by -1 composed with a cubic twist, and χ_{-1} is ramified only at 2 and ∞ .

9.3. And the converse holds

Finally, a theorem which says that Theorem A was not merely **an** explanation of the correlation but the **only** possible one.

Yu ([arXiv:1610.01195](https://arxiv.org/abs/1610.01195)). Let E, A be elliptic curves over a number field K . If

$$|\dim \text{Sel}_2(E^\chi) - \dim \text{Sel}_2(A^\chi)|$$

is bounded independently of χ , then there is a G_K -isomorphism $E[2] \cong A[2]$.

This settles, in the affirmative, the question the companion note opens with. The 2-Selmer ranks of $y^2 = x^3 + a$ and $y^2 = x^3 - a^{-1}$ track each other to within 1 across the whole twist family; by Yu's theorem that **forces** the mod-2 modules to be isomorphic. Theorem A exhibits the isomorphism explicitly, but its existence was already implied by the numerics.

10. Summary

what local conditions buy	because	flagship
descent is an algorithm	finitely many local questions (F1)	Bhargava–Shankar: mean $\#\text{Sel}_2 = 3$
Selmer groups can be compared	Greenberg–Wiles is a product over v	Cassels' isogeny formula; 2-descent

what local conditions buy	because	flagship
III can be bounded above	move one condition, use duality	Kolyvagin; Kato; Skinner–Urban
parity is unconditional	self-duality $\Rightarrow$ Lagrangians (F2)	p -parity theorem; BSD mod squares
Selmer statistics exist	random maximal isotropic model	Poonen–Rains; Smith; Goldfeld
objects can be defined by conditions	no Kummer map needed	Bloch–Kato; $R = \mathbb{T}$; level-raising
transfers between curves	functoriality (F3)	Selmer companions

Where the companion note lives. Rows 2, 4 and 7. Its Theorem A is a functoriality statement (F3); its Lemma B' is (F2) in the sharp form “the dimension of L_v is an invariant of the module”; its $(\star)$ is a companion-type comparison stronger than the parity consequence of Greenberg–Wiles that row 4 would give; and its one computational input is the local condition at $p = 2$ for $p = 2$, which is where the general theory also stops.

The one open question it suggests. Mazur–Rubin’s Theorem 3.1 needs potentially multiplicative reduction at 2 precisely because the Tate parametrisation is what makes the local condition at 2 legible. For a curve with $j = 0$ there is no such parametrisation, and the note’s answer at $v = 2$ is a finite enumeration. A criterion for 2-Selmer companionship at potentially **good** reduction — some CM-theoretic substitute for the canonical subgroup — would both prove the note’s theorem by hand and extend Theorem 3.1 in the direction its authors flag as the subtle one.

11. Reading list

- [1] B. Mazur, K. Rubin, **Kolyvagin systems**, Memoirs AMS **168** (2004). The reference for the Selmer structure formalism; Chapter 1 is the definition and the duality sequence of Section 3.3.
- [2] K. Rubin, **Euler systems**, Annals of Math. Studies **147** (2000). Local conditions as a technique, from the ground up; Chapter I is the best short account of (F1)–(F3).
- [3] H. Darmon, F. Diamond, R. Taylor, **Fermat’s Last Theorem**, §2. Greenberg–Wiles stated and proved, in the deformation-theoretic context that motivates it.
- [4] J. Neukirch, A. Schmidt, K. Wingberg, **Cohomology of Number Fields**, §8.6–8.7. Poitou–Tate and the general comparison formula, in the generality of Section 3.4.
- [5] B. Poonen, **Selmer group heuristics and sieves** (AWS 2014 notes). The Lagrangian picture of Section 6, informally.
- [6] B. Mazur, K. Rubin, **Selmer companion curves** (2012). The paper of Section 9.